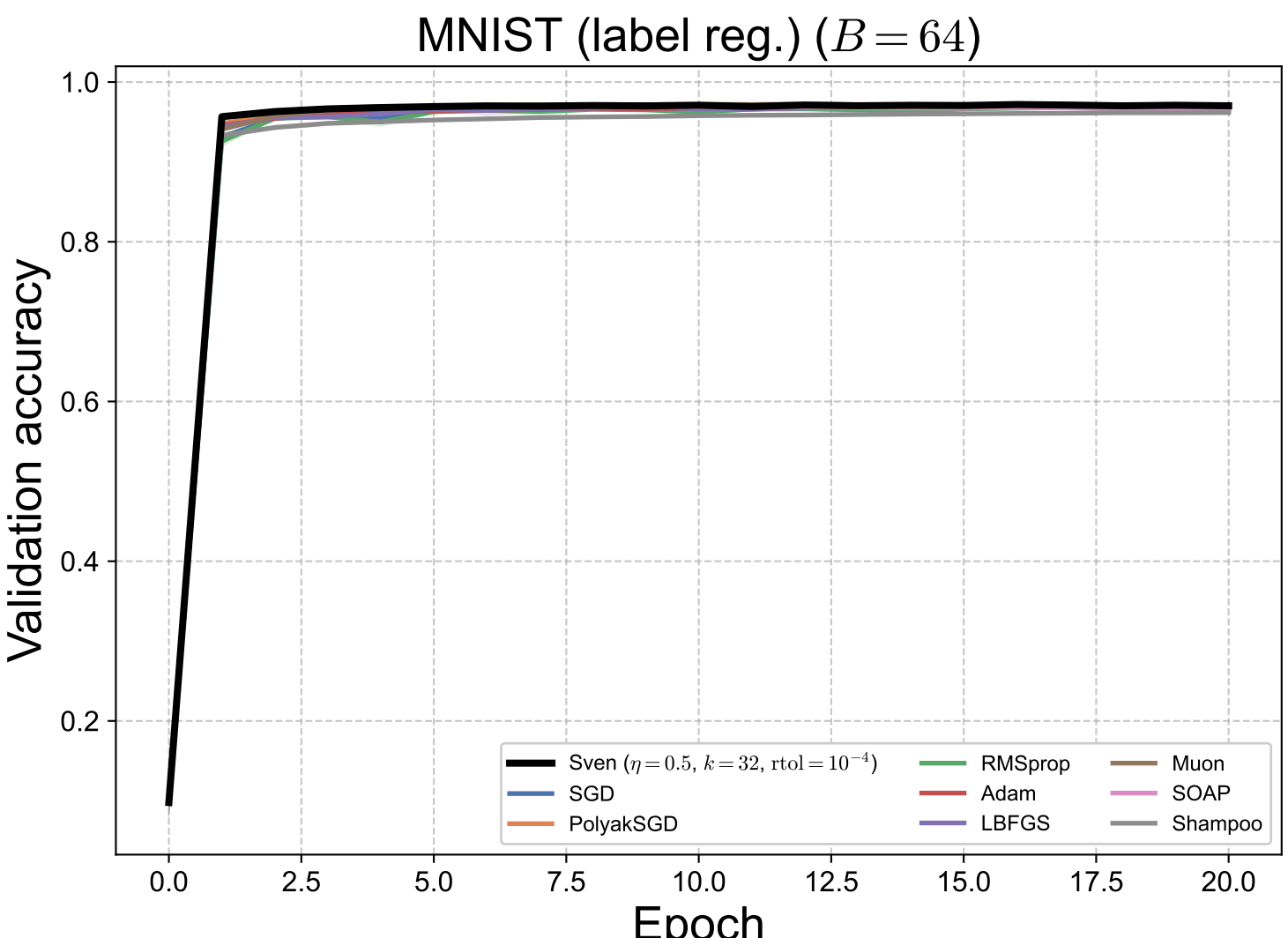

MNIST (label reg.) ($B = 64$)
Validation accuracy
Epoch
1.0
0.8
0.6
0.4
0.2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Sven ($\eta = 0.5$, $k = 32$, $\mathrm{rtol} = 10^{-4}$)
SGD
PolyakSGD
RMSprop
Adam
LBFGS
Muon
SOAP
Shampoo